

Highly accelerated life testing (HALT) of GaInNAs solar cells using temperature step-stress

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ABSTRACT

Dilute nitrides are proposed to form part of very efficient III-V multijunction solar cells. A key aspect for the development of dilute nitride solar cells is to determine their reliability. In this paper, we present for the first time a highly accelerated temperature step-stress life test with current injection on dilute nitride solar cells to emulate their electrical operation with the aim of determining their main reliability parameters. We have carried out a step-stress life test with increasing temperatures up to 230 °C. During the test, the cells were biased with the equivalent current they would photogenerate at about 50 suns, to emulate their performance in terrestrial medium concentration applications. For this purpose, we have manufactured GaInNAs single junction solar cells grown by Metal Organic Vapor Phase Epitaxy. The cells were properly assembled to withstand the high temperatures of the test. The accelerated life test lasted 1967 h. By assuming a failure criterion of an illumination power loss of 10%, just 4 solar cells failed of the 25 tested cells. The reliability analysis has been applied for two potential applications of these solar cells, namely terrestrial concentration and geosynchronous earth orbit space missions. Considering temperature stress and current injection, GaInNAs solar cells performed as robust devices with operation lifetimes much larger than the required ones. The results obtained here for GaInNAs solar cells can also be used as a general trend to guide the reliability analysis of dilute nitride optoelectronic devices.

1. Introduction

Dilute nitrides are proposed to be incorporated in III-V multijunction solar cells since several works demonstrate that such monolithic III-V solar cells with more than four junctions can achieve efficiencies over 50% under concentrated terrestrial sunlight, and over 35% for space operation [1–4]. Dilute nitride cells grown by Metal Organic Vapor Phase Epitaxy (MOVPE) have attracted the attention of the photovoltaic community to reduce solar cell fabrication cost since MOVPE is the growth technique of III-V solar cells [3,5] with the highest throughput.

As for any other type of upcoming solar cell, a key aspect for the development of dilute nitride solar cells is to determine their reliability [6]. There are many stressors affecting the performance and lifetime of solar cells, depending on their application. In terrestrial concentrator solar cells, high temperature, thermal cycling, UV light intensity, etc. greatly affect their performance evolution. The qualification standard

for concentrator solar cells which considers all these stressors is the IEC 62787 “Concentrator photovoltaic (CPV) solar cells and cell-on-carrier (CoC) assemblies - qualification” [7]. In the case of space solar cells, additional stressors such as radiation, atomic oxygen erosion, etc. must be considered such as in the European qualification standard ECSS-E-ST-20-08C [8] or the American AIAA S-111A counterpart [9]. In all these standards, testing the cells at high temperatures is compulsory. These standards provide a qualification criterion of “pass” or “not pass” of a product but they do not allow the estimation of their reliability parameters (estimate lifetime, Mean Time To Failure (MTTF), failure rate, etc). The procedure for estimating these reliability parameters is by means of Accelerated Life Tests (ALTs).

A previous high temperature storage test at 180 °C was performed with dilute nitride solar cells [10]. However, that work did not estimate the reliability parameters. Besides, the cells were not biased during the test which is compulsory to emulate the on-field working conditions by

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placing the devices at the nominal electrical conditions. Therefore, that work presented overestimated durability values. Nowadays there are no published works which estimate the main reliability parameters of dilute nitrides solar cells by means of temperature ALTs with current injection. Accordingly, in this paper we present a temperature Highly Accelerated Life Test (HALT) by emulating the nominal electrical operation of the solar cells with current injection, which results in the estimation of the lifetime, failure rate, and related reliability functions of these solar cells for both terrestrial concentration and space applications. Several methods exist for developing temperature Highly Accelerated Life Tests (HALT). In this work, a high-temperature step-stress test was conducted at the nominal bias current [6,11], as dilute nitrides solar cells are still in the development stage.

Because of the similitudes of the tested solar cells with other dilute nitride devices (semiconductor materials, structures, manufacturing technology), such as photodiodes, the results here shown can be of help to guide additional reliability analysis in dilute nitride optoelectronics.

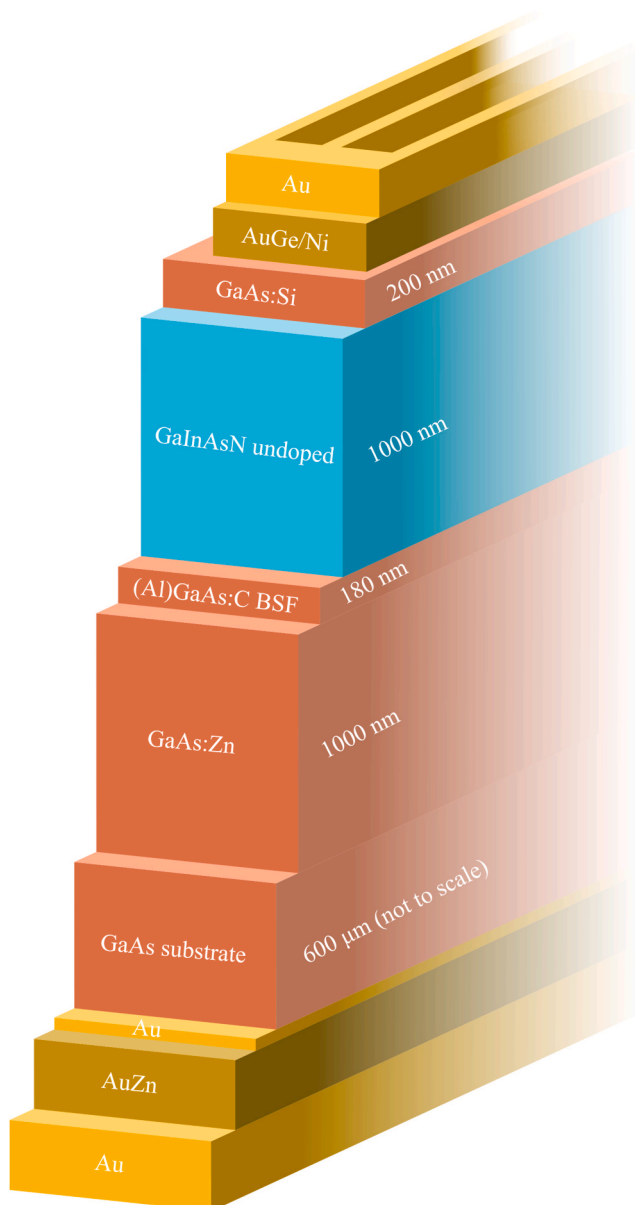


Fig. 1. Structure of the GaInNAs solar cells.

2. Manufacturing of GaInNAs solar cells

The semiconductor structures (see Fig. 1) were grown by MOVPE in a horizontal AIX200/4, Aixtron research-scale reactor. Note that the solar cell structure is of the *p-i-n* type, where the GaInNAs layer acts as the intrinsic zone. The introduction of nitrogen produces low diffusion length and lifetimes; therefore, this intrinsic region is introduced to improve collection efficiency due to the influence of the electric field produced [12]. The precursors used are AsH₃, PH₃ and TBHy for group V elements; TMGa, TMin, TMAI for group-III; and DMZn, DTBSI for the dopant elements.

After the epi-structure growth process, 1.4×1.5 mm (2.1 mm^2) active area devices are manufactured. Front contact is established by photolithography, followed by thermal evaporation of AuGe/Ni/Au, lift-off and Rapid Thermal Annealing (RTA). The back contact consisting of Au/AuZn/Au is subsequently deposited also by thermal evaporation and followed by RTA process. Both RTA processes were performed in a forming gas environment. Mesa isolation is performed via wet etching using a solution of ammonium hydroxide, hydrogen peroxide, and deionized water (see Fig. 2). Device individualization is accomplished through a manual scribing tool.

To ease handling of solar cells during the HALT, they are assembled on Direct Bonded Copper (DBC) substrates. These DBCs consist of a ceramic block with two 0.7 mm thick gold-plated copper plates separated by a 0.65 mm alumina channel. The back side of the solar cells is soldered to one of the metal plates on the DBC, while the front contact is connected to the other DBC metal plate by wire bonding. The back solder has a melting point of 237 °C [13]. For the front connection, four gold wires with a diameter of 25 μm are used. We assembled 27 devices like in Fig. 3.

3. HALT set-up

To carry out the HALT we develop a step-stress test type with bias current [14]. A step-stress test is a kind of HALT that consists of incrementing one stress parameter affecting the device in a stepwise fashion while the rest of parameters are kept at the nominal working values. In our case, the variable parameter (stressor) is the solar cell temperature. The main advantage of the step-stress test is that it allows to obtain reliability information in one unique test, reducing cost and time.

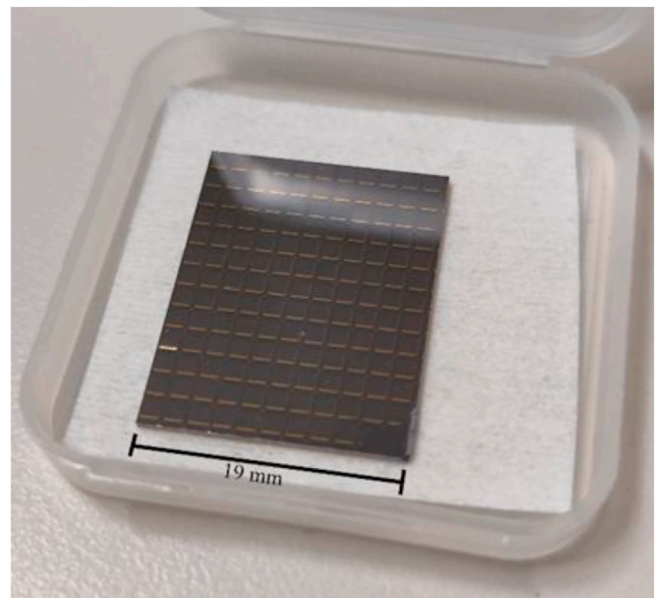


Fig. 2. Epiwafer piece containing tens of GaInNAs solar cells after mesa etching.

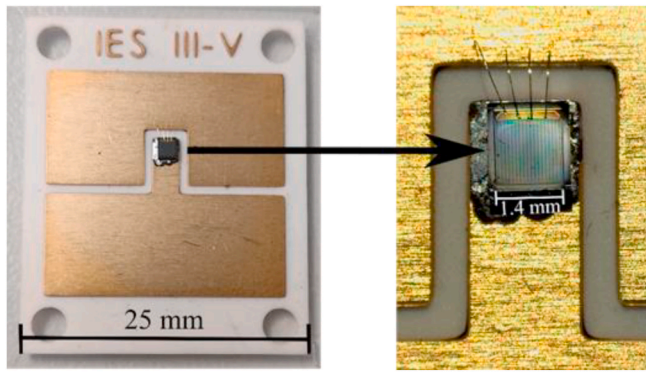


Fig. 3. Right: Top view of a GaInNAs solar cell after soldering and wire bonding on the DBC. Left: Whole view of a DBC with a GaInNAs solar cell.

Besides, after a step-stress test it is possible to know if the devices fulfil a reliability target, as well as their weaknesses from a reliability point of view. Therefore, step-stress tests are highly advisable when developing new devices with insufficient history and experience, such as in the case of dilute nitride solar cells. Moreover, step-stress tests allow to achieve quantitative reliability results, assuming that the activation energy of similar devices is known (which can be obtained with other HALT test types).

Dilute nitride solar cells can be used for operation under terrestrial concentration conditions or in space applications, so both situations are considered in this work. The electric working conditions can be emulated by forward biasing the solar cells at the nominal current injection level they would photogenerate at each specific application [15–19]. Of the aforementioned two applications, we have chosen the most demanding one in terms of operation current, i.e. terrestrial concentration which is several times (tens, hundreds ...) the operation at one sun [20]. Characterization before starting the HALT by means of dark I-V (DIV) curves shows that the effect of series resistance becomes noticeable from 0.7 to 0.9 A/cm². Therefore, we have chosen current densities below these values, namely 0.67 A/cm² (or 14 mA in terms of current), to be injected to the cells during HALT. This current density value is equivalent to the operation under 48 suns provided that these cells have an average $J_{sc} \sim 14$ mA/cm² (or 0.294 mA in terms of current) at 1 sun AM1.5d spectrum.

Considering that the nominal operating temperatures range from around 50 °C for concentrations at about 50 suns [21] to about 60 °C for geosynchronous Earth orbit (GEO), we decided to start the step-stress HALT at 80 °C in order to apply a certain acceleration factor. Subsequently, we increased the temperature steps to 100 °C, 130 °C, 160 °C, 190 °C, 203 °C, 215 °C, and 227 °C. This last temperature is adopted as the maximum one of the HALT to keep a security temperature margin regarding the melting point of the solder used in the solar cell assembly (237 °C). These temperatures increase approximately 3 °C due to the current injection applied to the cells during the HALT. For the sake of simplicity, this difference is omitted when naming temperatures throughout this paper, but it is taken into account in the calculation of reliability parameters in Section V. Thus, we say 80 °C instead of 83 °C and so on.

The method for calculating the real solar cell temperature is detailed elsewhere [22,23]. Regarding the selected spent time at each temperature step, it is important to note that a few days at 80 °C can be comparable in terms of degradation to, for example, a few hours at 190 °C due to the different accelerating factors of each temperature. Therefore, while conducting a step-stress test, if no significant degradation is observed at lower temperatures, it is advisable to go to a step at higher temperature in order to keep the HALT within an acceptable time duration.

A Thermotron SM-8-8200 climatic chamber has been used for the

HALT, where temperature can cycle between –70 °C and 180 °C [24]. To reach higher temperatures, the assembled cells of Fig. 3 were mounted on rectangular-shaped resistors that we called “heating carriers” from now on. To bond the assembled solar cells to the heating carriers we have used a silicone that withstands temperatures up to 300 °C, thereby not limiting the maximum temperature range of the HALT. To monitor the temperature of the heating carriers, thermocouples were inserted into them. Each heating carrier holds 5 assembled cells, 4 of which have current injection for all the steps, while the other one is only exposed to temperature but without current injection (see Fig. 4).

In this way, the HALT counts on 5 heating carriers holding 20 cells exposed to degradation from temperature with current injection, and 5 cells exposed only to temperature, without current injection. Comparing the evolution of both groups of solar cells during the HALT, allows us to segregate between the effect of current injection and that of the temperature on the degradation of solar cells. There are also 2 cells outside the chamber without any degradation, which are used as additional references of the initial stage (called as “ref cells”).

To track the evolution of the solar cells during the HALT, DIV measurements were carried out within the climatic chamber. The process begins by ensuring that each cell reaches and maintains the target temperature set for each step. A steady-state temperature and current injection are then established, with temperature measurements taken at regular intervals to ensure precise temperature control of each carrier. Every 6 h, the current injection stops, and DIV measurements are carried out again. All DIV measurements inside the climatic chamber are performed remotely. These measurements allow us to control the evolution of the cells without stopping the high temperature stress, avoiding unnecessary temperature cycling. Once the temperature step is completed since a pre-established safety time for the step is elapsed or we have observed a noticeable evolution of the dark I-V measurements, the cells are removed from the chamber, and external quantum efficiency (EQE), light I-V (LIV), and concentration I-V measurements are taken. After this characterization, the cells are placed back in the chamber where DIV measurements are conducted at 25 °C and 80 °C for comparison purposes among different steps. Afterwards, temperature increases to the next step's value, and the process is repeated. The summary of HALT steps is presented in Fig. 5.

We set as failure criterion a 10% decrease in the illumination power at the maximum power point of the solar cells, in accordance with similar reliability studies of solar cells [25,26].

4. Development of the HALT

During the HALT development, we detected three different stages in the evolution of the solar cells, depending on the temperature range. These three ranges are from 80 °C to 190 °C, from 190 °C to 227 °C, and at 227 °C.

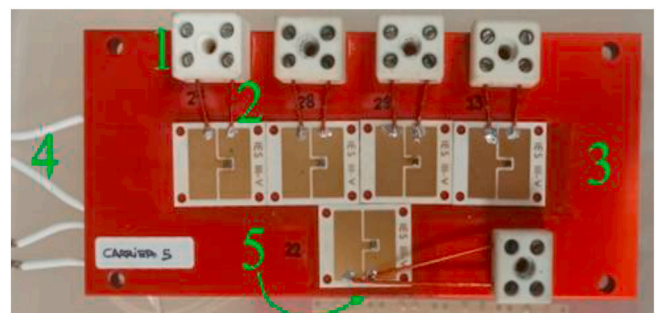


Fig. 4. Image of a heating carrier where: 1: terminal block, 2: copper wires, 3: rectangular shape resistance (heating carrier), 4: external wires to heat up the resistance, 5: location of the thermocouple.

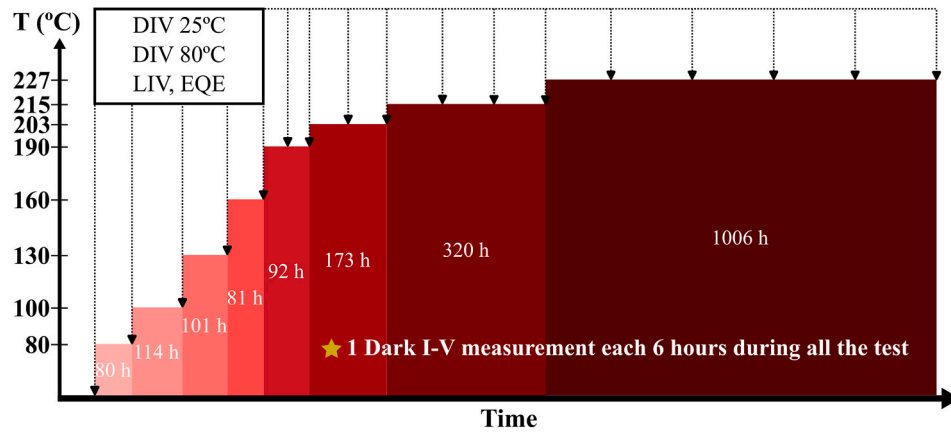


Fig. 5. Diagram of the HALT steps. Note that more than one characterization batch was performed in some temperature steps.

A. Evolution from 80 °C to 190 °C

Along the steps within this temperature range, we observe a shift of the dark I-V curves at high currents as consequence of a series resistance increase (see Fig. 6 left). Fig. 6 right shows the illumination I-V curves at concentrations where the solar cells photogenerate a J_{sc} of $\sim 0.6 - 0.7 \text{ A/cm}^2$, which is in the range of high currents of the dark I-V curves just before series resistance impacts their performance. As Fig. 6 right shows, solar cells at the end of the 160 °C step also exhibit a higher series resistance than the non-degraded cell ("ref cell", pink dashed-dot line in Fig. 6 right). No differences in the behavior between cells with current injection and without it were observed.

The evolution of series resistance in dark I-V curves is mainly due to vertical series resistances of the devices [27]. Accordingly, the increase of series resistance could be mainly explained by a degradation of the contacts of the cells, as observed in previous reliability works [16,26] but might be also due to an increase of the equivalent resistance of semiconductor heterojunctions. As the objective of this work is mainly to test the reliability of the dilute nitride semiconductor structure and not that of the contacts, we decide to apply the aforementioned failure criterion of 10% power loss at ~ 1.5 suns (instead of at 48 suns, in

concentration) since at this low illumination level the solar cells are unaffected by series resistance. This way, if we detect degradation of the solar cells at ~ 1.5 suns, it would be related with changes in the semiconductor structure. Since contact degradation could be another failure mode in terrestrial/space applications [28], it should be studied in future tests once the semiconductor structure performs as reliable.

B. Evolution from 190 °C to 227 °C

Contrary to what is observed in the previous steps (from 80 °C to the end of 160 °C), in the current steps the dark I-V curves shift to the left (see Fig. 7).

The cells show a higher recombination current (I_0) which results in a decrease of the open circuit voltage (V_{oc}) in the illumination I-V curve. Additionally, the illumination curve shows a significant rise in the short circuit current (Fig. 8).

The same short circuit current increase is derived from the EQE measurements. In order to confirm that this J_{sc} increase derives from an improvement of the quality of the dilute nitride and not from the change of the optical properties of the surface of the cell due to temperature, we measure the reflectivity of the cells, which remains constant. So, the

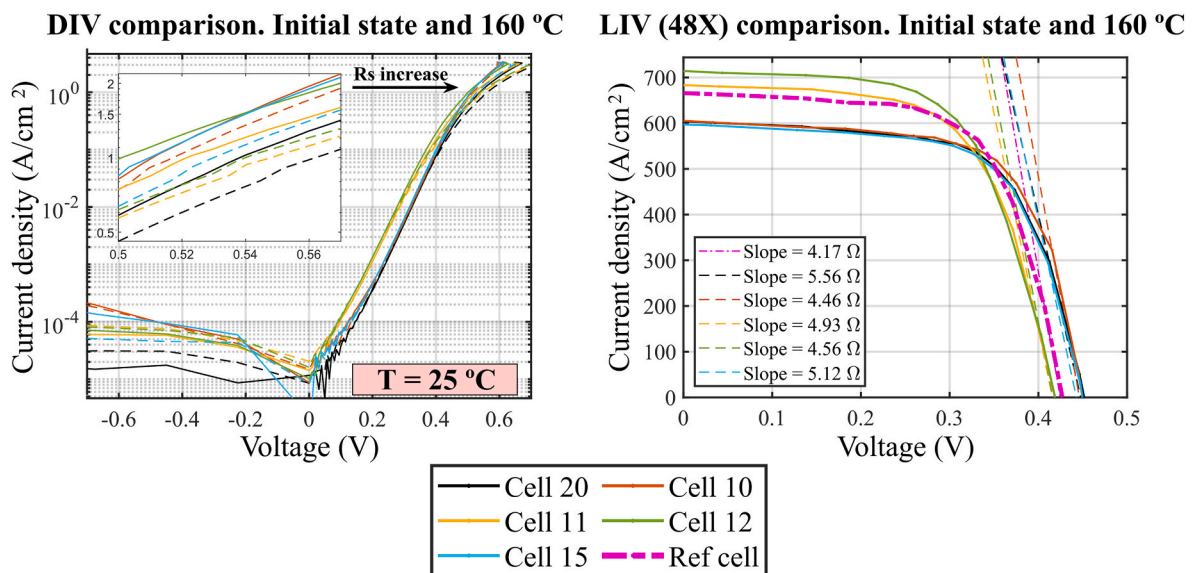


Fig. 6. Left: Evolution of the dark I-V curves of several cells with current injection. Solid lines represent the dark I-V curves at the initial stage before starting the HALT, while the dashed lines are the I-V curves after the 160 °C step. All measures are taken at 25 °C. Right: Comparison of I-V illumination curves of cells after 160 °C step stress versus a ref cell at 48 suns. Linear fittings of the last points of the curves have been made to estimate the series resistance of the cells, where the increase in resistance is observed.

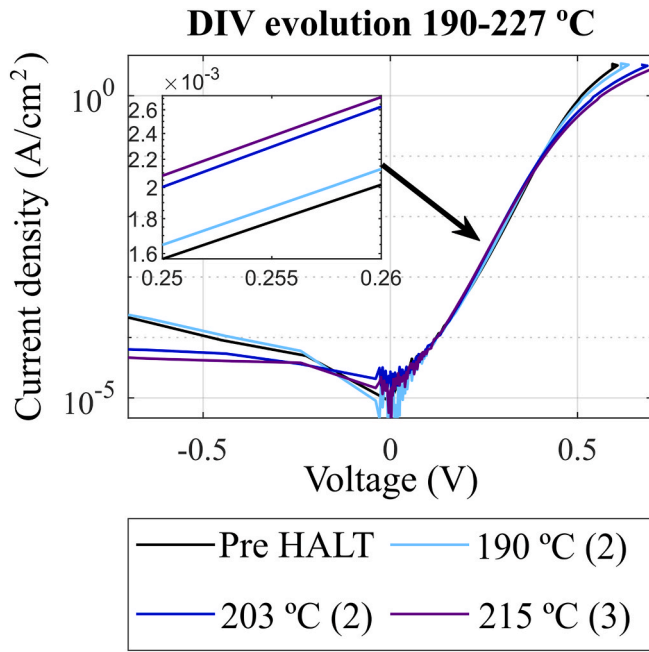


Fig. 7. DIV evolution of a representative solar cell between Pre Alt and during high temperature steps up to 215 °C step (just before 227 °C). The numbers in brackets of the legend indicate the characterization stage out of chamber for the same temperature step (for clarity and to show tendency, we are presenting only second characterization at 190 °C and 203 °C and the third characterization at 215 °C). All measurements are taken at 25 °C.

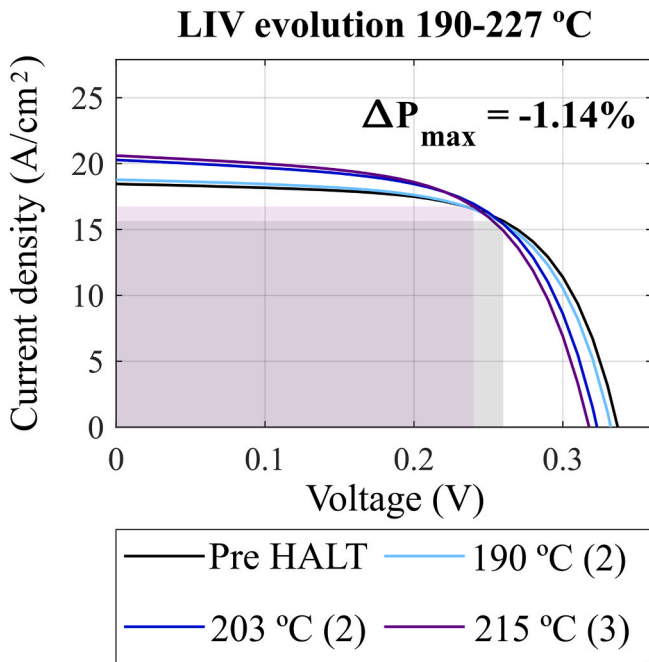


Fig. 8. LIV evolution of the representative solar cell of Fig. 7 between Pre Alt and during high temperature steps up to 215 °C step (just before 227 °C). All measurements are taken at 25 °C. The power loss shown has been calculated as the relative difference between the maximum power point for the pre-HALT measurement and the final characterization at the end of 215 °C step.

increase of EQE and J_{sc} is attributable to the semiconductor structure. This EQE increase tends to stagnate with temperature (see Fig. 9).

The increase of J_{sc} compensates in a large extent the V_{oc} decrease, so the illumination power loss does not reach the failure criterion of -10%

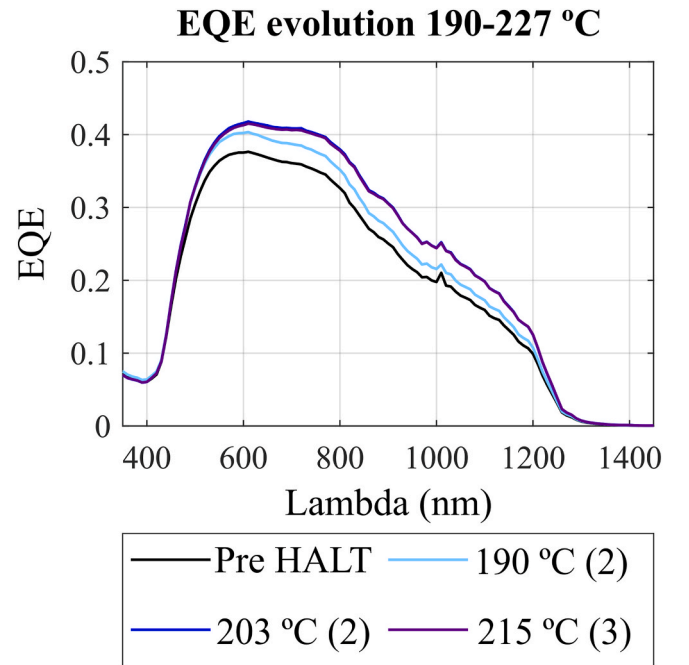


Fig. 9. EQE evolution of the representative solar cell of Fig. 7 between Pre Alt and during high temperature steps up to 215 °C step (just before 227 °C). All measurements are taken at 25 °C. Note that the EQE increase stagnate at 203 °C.

in any solar cell (as shown in Fig. 8). Even some cells experienced an increase of illumination power. In the end, no solar cell can be considered as failed. Just like before, no differences in behavior were observed between cells with current injection and without, indicating that the current injection level we apply is not high enough to promote degradation.

C. Evolution of the solar cells at 227 °C

As previously mentioned, 227 °C is the maximum temperature we can use in the HALT due to the maximum temperature tolerance of the solder used in assembling the solar cells. Therefore, we kept the solar cells at this temperature until the end of the HALT. The total time the cells were maintained in this step was about 1000 h (more than 50% of the total HALT duration) and during this time they were taken out of the climatic chamber five times for illumination and EQE measurements.

Regarding the variation of the dark I-V curves, a gradual shift of the curves to the left is observed (see Fig. 10).

This increase in I_0 results in a V_{oc} drop (Fig. 11).

Additionally, the EQE increase which tended to stagnate at temperatures below 227 °C, now changes and EQE experiences a significant increase (Fig. 12).

However, the origin of EQE rising does not seem to be similar as in previous temperature steps where the increase in EQE was for the whole wavelength range (see Fig. 9). Now the EQE increase is more localized in the long wavelength region (see Fig. 12). Whereas in previous steps, the increase in J_{sc} overcompensated the V_{oc} losses, this is not what we see at this final HALT step. The reason is the appearance of a low parallel resistance-like effect, which is not observed in the previous temperature steps. This decrease of parallel resistance could be related to the degradation of the GaInNAs material quality or a reduction in the background doping in the intrinsic layer [12,29]. In fact, the parallel resistance-like effect in dilute nitride solar cells has been attributed to electric field-aided collection [30]. This parallel resistance reduction results in a fill factor (FF) decrease, and therefore, the illumination power of the tested cells worsens significantly (see Fig. 11).

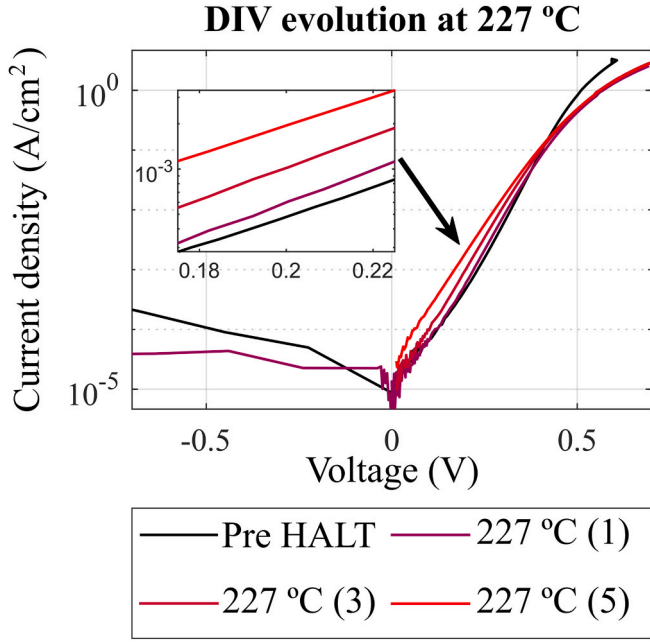


Fig. 10. DIV evolution of the representative solar cell of Fig. 7 during 227 °C temperature step. All measurements are taken at 25 °C.

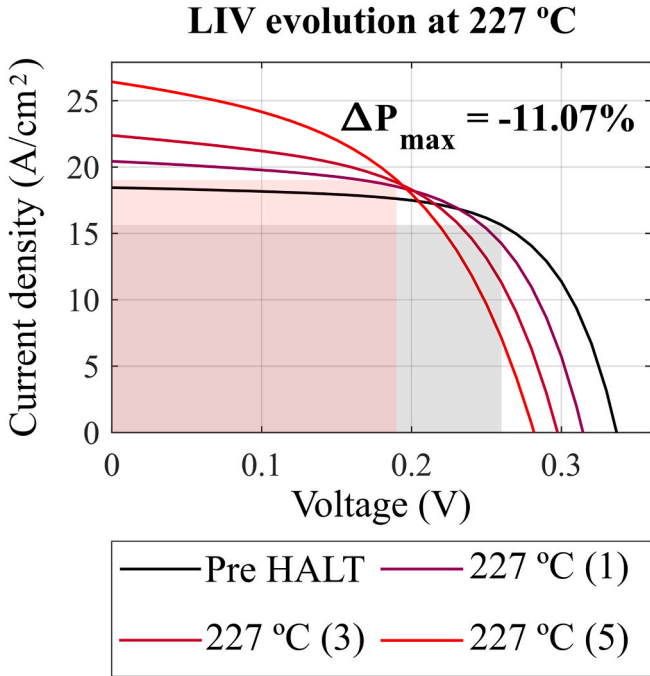


Fig. 11. LIV evolution of the representative solar cell of Fig. 7 during 227 °C temperature step. All measurements are taken at 25 °C. The power loss shown has been calculated as the relative difference between the maximum power point for the pre HALT measurement and the fifth measurement at the end of the step at 227 °C.

At the end of this step, the cells that have achieved the highest EQE increase were, at the same time, the ones with both the highest V_{oc} and FF drops, leading to the highest illumination power loss. Four cells surpassed the failure criterion of 10% illumination power loss and they are considered as failed cells. All cells showed power losses with similar performance degradation, so we can conclude that current injection at the levels used in the HALT is not a factor that increases the degradation

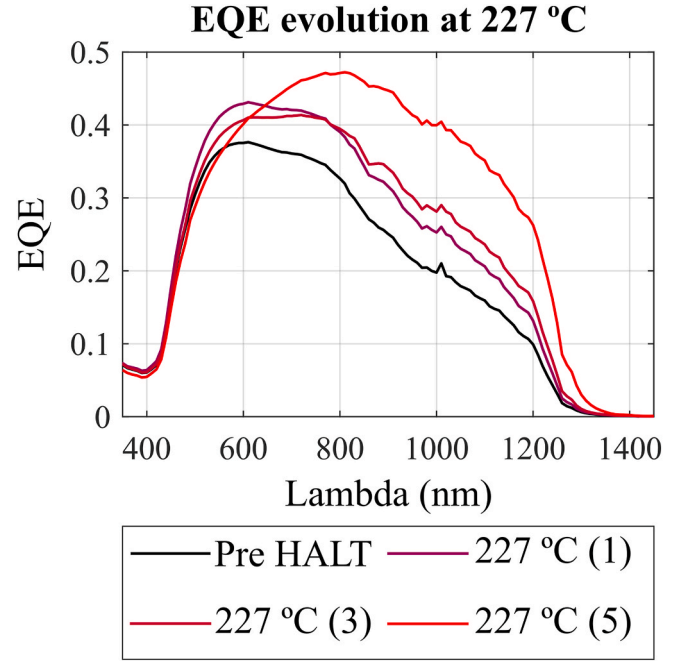


Fig. 12. EQE evolution of the representative solar cell of Fig. 7 during 227 °C temperature step. All measurements are taken at 25 °C.

of the devices.

5. Estimation of reliability parameters

To conduct the reliability assessment once the HALT has finished, it is crucial to determine the time elapsed to each failure [31]. However, in terms of degradation, the same elapsed time at different temperatures does not represent the same degradation time. For this reason, the elapsed time at each temperature step must be normalized to an equivalent time at a given temperature. The Arrhenius model allows this normalization at the target temperature [32], using (1). As we anticipated in Section 3, now we will include the 3 °C increase in the solar cell temperature as consequence of the current injection. Therefore, the noted temperature is 230 °C (instead of 227 °C), which is the solar cell temperature where all the failures occurred. The normalization of elapsed times at different temperature steps is achieved by using the acceleration factor, AF :

$$AF_{T_{step} \& T_{target}} = t_{target} / t_{step} = \exp(E_a / k_b (1 / T_{step} - 1 / T_{target})) \quad (1)$$

where.

- $AF_{T_{step} \& T_{target}}$ = Acceleration Factor between the step temperature and the target temperature.
- T_{step} : temperature of the step (K).
- t_{step} : elapsed time hours at T_{step} . Known value during test.
- t_{target} : equivalent time (hours) at the target temperature, estimated according to (1).
- T_{target} : target temperature (K) to normalized times, i.e. 230 °C, in this case.
- k_B (eV/K): Boltzman's constant.
- E_a (eV): activation energy of the degradation/failure process.

Since there are not previous HALT works on GaInNAs solar cells, there are not available data of the activation energy. Notwithstanding the above, we use as a first approach an activation energy of 0.5 eV which serves as a rounded intermediate number between the 0.41 eV

defect level reported in GaInNAs solar cells grown by MBE [33] and the 0.667 eV defect level observed in dilute nitride solar cells grown by MOVPE [34], as these are the most comparable study cases available. Although we will initially use 0.5 eV as the starting point, we will evaluate how variations in activation energy values impact the reliability results at the end of this section.

By using (1), AF experienced by the solar cells at each temperature step is calculated with respect to 230 °C as it has been applied previously [19]. This allows the calculation of the equivalent duration of all steps (the whole HALT) at 230 °C, resulting in 1404 h (see Table 1).

During the test, four solar cells failed when applying the failure criterion of a 10% power decrease. These four solar cells failed at an accumulated time of 897, 1045, 1211 and 1306 h, respectively. With these values, we carry out a reliability analysis using the HBK ReliaSoft (Reliability Analysis and Management) software tool and its ALTA Module (Accelerated Life Testing Data Analysis). Failure times are fitted to a two-parameters Weibull failure distribution function, namely β –shape parameter– and η –scale parameter– due to its versatility and its demonstrated ability to provide excellent fits in previous temperature ALTs carried out on III-V concentrator solar cells [16]. The selected fitting method is Maximum Likelihood Estimation (MLE) using the Standard Ranking Method (SRM) to assess the cumulative failure probability. Likelihood Ratio Method has been used to estimate the Confidence Bounds [35]. Once Weibull parameters, β and η , have been assessed, the reliability function, $R(t)$, which is the probability that solar cells work properly at a given time, can be calculated:

$$R(t) = \exp \left(- (t/\eta)^\beta \right) \quad (2)$$

where β indicates the life stage of the solar cell as follows: $\beta < 1$ means an infant stage, $\beta = 1$ denotes a mature stage and $\beta > 1$ an aging stage. The scale parameter η indicates the time at which reliability function is $e^{-1} = 0.368$.

Fig. 13 presents the failure times of the four failed cells in the Weibull plot, showing that the model with the maximum likelihood estimation (MLE) method fits very well the experimental points. The values of the parameters obtained after fitting to the Weibull model at 230 °C are $\beta = 4.53$ and $\eta = 1840$ h. A $\beta > 1$ (as in this case) implies that the solar cells have a failure rate that increases with time, which corresponds to a degradation stage (wear out period of life that corresponds with failure rate increasing with time).

With the assessed β and η parameters it is possible to evaluate the reliability at any instant of time by means of equation (2). From this reliability analysis, some additional key reliability parameters at 230 °C are assessed, such as MTTF (Mean Time to Failure) = 1680 h and the time at which $R(t) = 0.9$, which results in 1120 h being the two-sided 90% confidence values 1382 h (right red line in Fig. 13-upper bound) and 747 h (left red line in Fig. 13-lower bound). The two side confidence values are directly related with the sample size and the number of failures. The obtained range is good enough for a preliminary test in this innovative solar cells.

Once the reliability parameters at 230 °C are assessed, we can evaluate the reliability at nominal working conditions. For the two

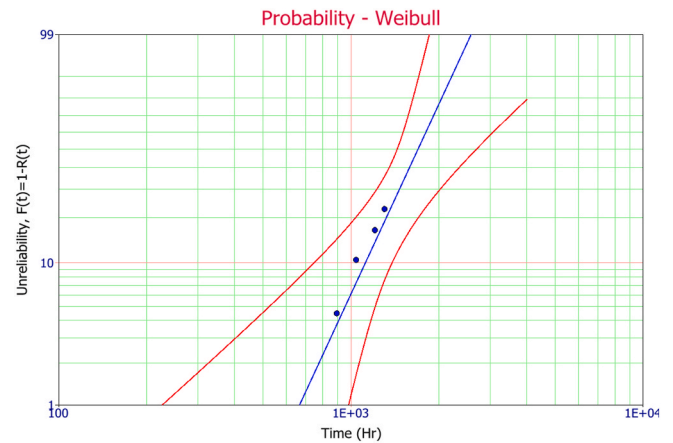


Fig. 13. Unreliability vs. time Weibull plot at 230 °C. The blue line is the best fitted unreliability function, $F(t) = 1 - R(t)$, and the red lines are the $F(t)$. two-sided 90% confidence values. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

applications we are considering, the calculated AFs by means of equation (1) are.

- Terrestrial concentration applications (CPV). For 48 suns we assume a solar cell temperature of 50 °C which results in $AF = 616$ (at 230 °C) over nominal temperature and an average working time of 10 h/day [36].
- Space applications (GEO). We assume a solar cell average temperature of 60 °C (although there are few eclipses per year in which temperature can range from –200 °C to 95 °C) which results in $AF = 360$ (230 °C) over nominal temperature and 20 h/day of working time [37].

By applying these AFs and hours/day of working time to both applications, their unreliability Weibull distributions are presented in Fig. 14 left. As can be seen, the time for the same reliability value is 3.4 times larger for CPV than for GEO due to lower working temperature in CPV. Based on the Weibull parameters for GEO and CPV, additional key reliability functions are assessed such as the failure probability density function, $f(t)$, that indicates the likelihood of failure for any time, and it describes the shape of the failure distribution. Fig. 14 right shows that failures in GEO (black line) occur earlier and in a more concentrated way than for CPV (blue line).

Fig. 15 left shows the reliability function where the four failure cells of the HALT are extrapolated to the nominal temperature of both applications. Obviously, CPV has a higher reliability than GEO since CPV has a lower temperature of operation. Fig. 15 right shows the failure rate function defined as the $f(t)/R(t)$ ratio.

Table 2 summarizes the main reliability parameters for both applications, CPV and GEO. As can be seen, the expected lifetime of the tested GaInNAs solar cells is much beyond the standard requirements. It must be noted that these values are obtained by considering the high temperature as the unique stressor. Obviously, the addition of other stressors as radiation, thermal cycling, etc. will decrease the time values. Therefore, in order to estimate the meaningful reliability figures of solar cells, a set of accelerating aging tests which include all these stressors is compulsory.

Given that in this section we have previously assumed an activation energy, $E_a = 0.5$ eV, and that this value could be different for the tested solar cells, an analysis on the influence of the activation energy on the reliability parameters is in order. Accordingly, Table 3 presents the same reliability parameters of Table 2 at GEO conditions but now also considering activation energies values of 0.4 eV and 0.6 eV. As can be seen, the influence of the activation energy is very high, but in the worst

Table 1

Conversion of all temperature step times to the equivalent time at 230 °C.

T_{step} (°C)	t_{step} (h)	AF with respect to 230 °C	Equivalent step time at 230 °C (h)	Accumulated time at 230 °C (h)
83	80	0.00854	0.68	0.7
103	114	0.02032	2.32	3.0
133	101	0.06354	6.42	9.4
163	81	0.16987	13.76	23.2
193	92	0.40014	36.81	60.0
206	173	0.56102	97.06	157.0
218	320	0.75432	241.38	398.4
230	1006	1	1006	1404.4

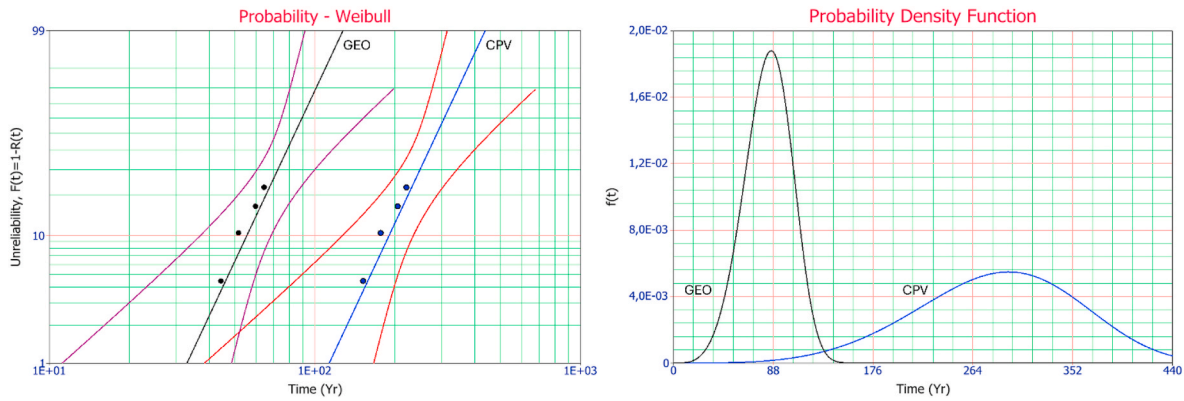


Fig. 14. Left: weibull unreliability with two-sided 90% confidence values. Right: probability density function $f(t)$ for GEO (black) and terrestrial CPV (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

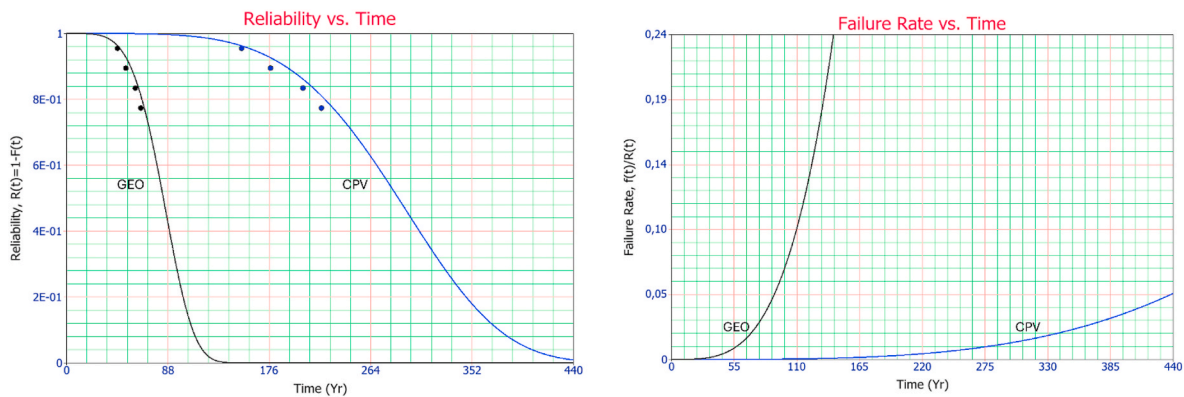


Fig. 15. Left: reliability function. Right: failure rate function for GEO (black) and terrestrial CPV (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

Main reliability parameters for both applications.

Application	β	η (yr)	MTTF (yr)	$F(10\%) = R(90\%)$ (yr)
GEO	4.53	91.0	83.1	55.4
CPV	4.53	312.2	285.1	190.1

case of $E_a = 0.4$ eV, the lifetime values still surpass the standard requirements. One important consideration is that our chosen activation energy range (0.4–0.6 eV) is considerably more pessimistic than the value assumed in Ref. [10], where an activation energy of 1.0 eV was taken. This higher value of 1.0 eV is characteristic of high-quality solar cells, where contact metal diffusion into the semiconductor is the primary failure mechanism. Therefore, the reliability values obtained in this study, which are lower than those reported in Ref. [10], are conservative values and likely more representative of the true performance of MOVPE GaInNAs solar cells.

Finally, we note that these results can be used as a general trend for the reliability of MOVPE grown GaInNAs optoelectronic devices. Considering that the typical operating temperatures of optoelectronic devices are near ambient conditions, MOVPE-grown GaInNAs

Table 3

E_a influence on the main reliability parameters for GEO conditions.

E_a (eV)	β	η (yr)	MTTF (yr)	$F(10\%) = R(90\%)$ (yr)
0.4	4.53	28.0	25.5	17.0
0.5	4.53	91.0	83.1	55.4
0.6	4.53	293.7	268.1	178.8

optoelectronic devices operating at current densities equal to or below those used in this study are anticipated to demonstrate reliable performance and sustained functionality over extended periods of operation.

6. Summary and conclusions

We present for the first time a long period highly accelerating life test in temperature on dilute nitride solar cells with bias current to emulate their electric nominal conditions and with the aim of determining their main reliability parameters. We have carried out a step stress HALT with increasing temperatures from 80 °C to 230 °C. During the test, the cells were forward biased with the equivalent current they would photo-generate at about 50 suns, to emulate their performance in medium concentration applications. For this purpose, we have manufactured GaInNAs solar cells grown by MOVPE. The cells were properly assembled to withstand the high temperatures of the test. The HALT lasts 1967 h.

By assuming a power loss of 10% as the failure criterion, just four solar cells failed of the 25 tested cells. The reliability analysis has been applied to two potential applications, namely terrestrial CPV and GEO space missions. By assuming an activation energy of defects of 0.5 eV, an MTTF (Mean Time To Failure) of 83.1 years is estimated for GEO while 285.1 years is obtained for CPV. This huge difference for both applications derives from their different nominal temperatures, which is 50 °C for CPV while 60 °C for GEO, and also for their nominal use time which is 10 h/day for CPV and 20 h/day in GEO. A complete set of reliability parameters such as the reliability function, failure probability density function, failure rate function, etc. have been also calculated. It must be noted that these reliability results are obtained by considering a high

temperature as the unique stressor. The addition of other stressors such radiation, thermal cycling, UV, etc. will decrease MTTF and the rest of the reliability parameters. Therefore, in order to estimate the meaningful reliability figures of GaInNAs solar cells a set of accelerating life tests which include all these stressors is compulsory.

Finally, the good reliability results achieved with these MOVPE grown GaInNAs solar cells can be also used as a general trend to guide the reliability determination of similar dilute nitride optoelectronic devices.

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CRediT authorship contribution statement

Alvaro Pulido-Muñoz: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Neftalí Núñez:** Writing – review & editing, Software, Methodology. **Manuel Vázquez:** Writing – review & editing, Software, Methodology. **Daniel Gómez-Reboreda:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Pablo F. Palacios:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Francisco Javier Jiménez:** Software. **Jesús Bautista:** Writing – review & editing, Methodology. **Luis Cifuentes:** Writing – review & editing, Methodology, Investigation. **Iván García:** Writing – review & editing, Supervision, Methodology. **Mercedes Gabás:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Carlos Algora:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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